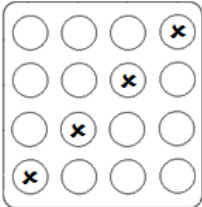


| Question |     |      |  | Marking details                                                                                                                                                                                                                                                                                                                                                                                                                               | Marks available |     |     |       |       |      |
|----------|-----|------|--|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------|-----|-----|-------|-------|------|
|          |     |      |  |                                                                                                                                                                                                                                                                                                                                                                                                                                               | AO1             | AO2 | AO3 | Total | Maths | Prac |
|          | (b) | (i)  |  |  <p>all four crosses shown (1)</p> <p>award (1) for any of following</p> <p>metals do not displace themselves from solution<br/>metals do not react with their own nitrate<br/>metals do not react with themselves</p> <p>metals have the same reactivity – neutral answer</p> <p>correct explanation may be credited when the first mark is not awarded</p> | 2               |     |     | 2     |       | 2    |
|          |     | (ii) |  | <p>0.381g / 0.38g / 0.4g (3)</p> <p>if answer incorrect credit working</p> <p><math>112 \leftrightarrow 190.5 / (2 \times 56) \leftrightarrow (3 \times 63.5) / \frac{0.224}{112}</math> (1)</p> <p><math>\frac{0.224}{112} \times 190.5</math> (1)</p> <p>alternative method</p> <p>0.004 mol Fe (1)<br/>0.006 mol Cu (1)</p> <p>ecf possible</p>                                                                                            |                 | 3   |     | 3     | 3     |      |